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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
Euthanasia Agents: Briefly, mice were anesthetized using ketaminexylazine (100 mg15 mgkg i.p.) and received a 65-kilodyne mid-thoracic (T8) contusion injury.
IACUC: All procedures were in accordance with University of Miami Institutional Animal Care and Use Committee and National Institutes of Health guidelines.
Inclusion and Exclusion Criteria
After selecting for viable CD45+ leukocytes , neutrophils were excluded based on Ly6G expression .
Attrition
not detected.
Sex as a biological variable
Materials and methods Mice and SCIs For sequencing, flow cytometry, and histological validation of tip cells, female C57BL6J mice 8-10 wk of age were purchased from Jackson Laboratory (stock 000664).
Subject Demographics
Age: Materials and methods Mice and SCIs For sequencing, flow cytometry, and histological validation of tip cells, female C57BL6J mice 8-10 wk of age were purchased from Jackson Laboratory (stock 000664).
Randomization
For each of 1,000 permutations, we randomly shuffled the cell-type and time point labels and calculated an interaction scores for all possible <ligand cell, receptor cell, time point> combinations.
Blinding
not detected.
Power Analysis
not detected.
Replication

We enriched our second set (and third set for uninjured and 1-dpi samples) of replicates with astrocytes by pooling from two spinal cord samples.	
Data Information	
Identifiers: , Zombie Aqua Dye , CD45.2 , CD11b , and Ly6G were the same as above P2RY12 (PE Biolegend 8480032721644 1:100) and MSR1 (Alexa Fluor 700 BioRad MCA1322A7002251086 1:50) .	P2RY12

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
Primary antibodies used were: chicken anti-GFAP (Abcam ab4674AB-304558 1:500) , rat anti-GFAP (Life Technologies 13-0300AB-86543 1:500) , rabbit anti-PDGFRb (Abcam ab32570AB-777165 1:200) , goat anti-OSMR (R&D Systems AF-662-SPAB-355511 1:100) , goat anti-Podocalyxin (R&D Systems AF1556AB-354858 1:200) , rabbit anti-APOE (Abcam ab183597AB-2832971 1:100) , CD63 (PEDazzle 594 Biolegend 1439132565503 1:100) , and rat anti-CD11b (BioRad MCA74GAAB-324660 1:500) .	anti-GFAP		
	anti-PDGFRb		
	anti-OSMR		
	anti-Podocalyxin		
	anti-APOE		
	CD63		
	anti-CD11b		
After centrifugation at 300 g for 10 min , supernatant was aspirated , and the cells were incubated in blocking buffer mixture for 5 min at 4degC (1 ml anti-mouse CD1632 TruStain Fcx RRID: AB-1574975 Biolegend cat 101320 and 99 ml FACS buffer 1 BSA + 0.05 sodium azide in HBSS wo Ca2+Mg2+) .	anti-mouse CD1632 TruStain Fcx	BioLegend	(BioLegend Cat# 101320, RRID:AB_1574975)(link)
Experimental Models: Organisms/Strains			
Materials and methods Mice and SCIs For sequencing, flow cytometry, and histological validation of tip cells, female C57BL6J mice 8-10 wk of age were purchased from Jackson Laboratory (stock 000664).	C57BL6J		Suggestion: RRID:IMSR_JAX:000664(link)
For histological validation of activated fibroblasts, Postn-CreER mice (Jackson Laboratory stock 029645) were bred to Rosa26-EYFP mice (Jackson Laboratory stock 006148) to generate PostnEYFP mice.	Postn-CreER		
	Rosa26-EYFP		
All PostnEYFP mice were of the C57BL6 background.	C57BL6		Suggestion: RRID:IMSR_JAX:000664(link)
Software and Algorithms			
Individual particle areas were restricted to an area of 2 um2 or larger using the Analyze Particles function in ImageJ .	ImageJ		Suggestion: ImageJ, RRID:SCR_003070(link)
In brief, base call files for each sample were demultiplexed into	STAR		

FASTQ reads and then aligned to the mouse mm10 reference genome using the STAR splice-aware aligner.			
To further distinguish cells from data below the knee, we used the emptyDrops function from the DropletUtils R package (Lun et al., 2019) using the following fixed parameters for all samples: lower = 250 maximum fit.bounds = 1e06 false discovery rate = 0.001 ignore = 10.	DropletUtils		
To remove potential doublets, we applied the Scrublet Python package (Wolock et al., 2019) for each individual sample.	Python		
In brief, the SingleR package performs a Spearman rank correlation between each query cell and assigned cell-type labels of a reference dataset.	SingleR		
To identify DEGs for GO analysis, we used the MAST framework as implemented in the Seurat FindMarkers() function and set the latent.vars argument to nFeature_RNA, which is equivalent to adjusting for the cellular detection rate (Finak et al., 2015).	MAST		
For GO analysis, we used Fishers exact test as implemented in the topGO package (Rahnenfuhrer, 2020).	topGO		
We first pulled a reference list of human ligand-receptor pairs published previously (Ramilowski et al., 2015) and converted the genes into mouse orthologues using the Ensembl biomaRt package (Durinck et al., 2009).	Ensembl		

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
In brief, the SingleR package performs a Spearman rank correlation between each query cell and assigned cell-type labels of a reference dataset.	Spearman rank correlation
Differential expression testing and GO To identify marker genes for each cluster, we used the FindAllMarkers() function using default parameters, which implements a Wilcoxon rank-sum test comparing gene expression of cells within a given cluster versus all other cells.	Wilcoxon rank-sum test
For GO analysis, we used Fishers exact test as implemented in the topGO package (Rahnenfuhrer, 2020).	Fishers exact test

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For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq